Increasing Response Diversity in LLMs via On-Policy Unlearning

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Abstract

Large language models (LLMs) often exhibit mode collapse in open-ended generation, repeatedly producing similar responses despite the availability of diverse valid alternatives. Existing approaches address this problem through diversity-aware optimization or decoding-time interventions, often requiring curated preference data, candidate selection, or additional inference. We adopt *on-policy unlearning*, a general framework that dynamically constructs unlearning targets from the model's current policy, and introduce UNMODE to mitigate response mode collapse. At each round, UNMODE samples responses from the current model and updates its parameters to reduce the probability of generating them. Resampling after each update lets the forget targets track the evolving policy, progressively suppressing likely responses and making alternatives relatively more probable. This approach requires neither curated replacement data nor separately sampled alternative responses. We evaluate UNMODE on two open-ended generation benchmarks and code-completion tasks. Results show that UNMODE improves response diversity and utility over diversity-oriented baselines. It also improves pass@5 and pass@10 on code completion, increasing the likelihood of finding correct solutions when sampling multiple candidates. These findings demonstrate the potential of on-policy unlearning as a simple and data-efficient approach to reducing response mode collapse in LLMs.

1 Introduction

Large language models (LLMs) are increasingly used for tasks without a single canonical answer, including brainstorming, creative writing, and exploring alternative solution strategies. In such settings, usefulness depends not only on the quality of an individual response but also on the ability to generate meaningfully different alternatives (Padmakumar and He, 2024; Zhang et al., 2025b). Diversity is also important for exploration-based learning, where repeatedly sampling similar trajectories can limit the discovery of successful solutions (Chen et al., 2025; Cheng et al., 2025; Cui et al., 2025; Li et al., 2025; Hu et al., 2026b).

Post-training can substantially narrow the output distribution of LLMs (GX-Chen et al., 2025; Padmakumar and He, 2024). Instruction-tuned and aligned models often exhibit *mode collapse*, repeatedly producing similar responses even when many valid alternatives exist (Slocum et al., 2025; O'Mahony et al., 2024; Kruszewski et al., 2025). This occurs both within models and across model families: Infinity-Chat reports substantial intra-model repetition and inter-model homogeneity on open-ended queries (Jiang et al., 2025), and NoveltyBench shows that state-of-the-art models generate far fewer distinct responses than humans (Zhang et al., 2025b). These findings reveal a trade-off in current post-training: optimizing for the most likely response can reduce the model's ability to explore alternative useful behaviors (GX-Chen et al., 2025; Kruszewski et al., 2025).

Existing efforts to recover diversity follow two directions. The first modifies the training objective: DivPO (Lanchantin et al., 2025) shifts preference optimization toward rare but high-quality responses, DARLING (Li et al., 2025) rewards semantic diversity during reinforcement learning, and Li et al. (2024) formulate diversity preservation during supervised fine-tuning as an adversarial game. The second leaves parameters unchanged and modifies inference, e.g., in-context regeneration (Zhang et al., 2025b) and Verbalized Sampling (Zhang et al., 2025a). Both have limitations. DivPO and DARLING require sampling and selecting diverse responses, which introduces a costly search process. Li et al. (2024) is specific to training time, making it expensive to apply to already-trained models, and requires access to the original training data. Inference-time methods increase inference

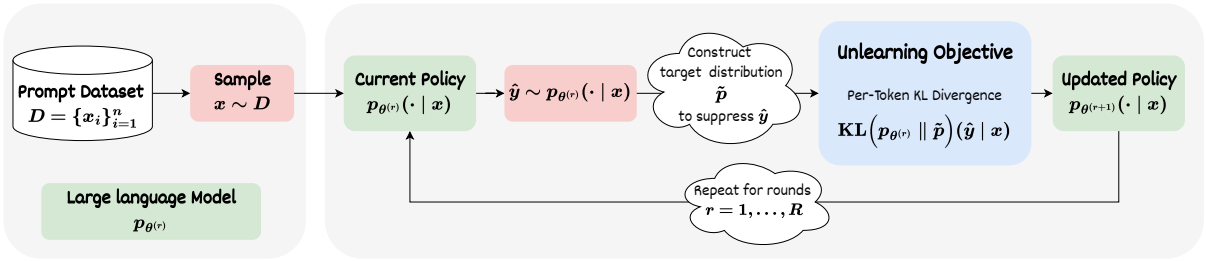


Figure 1: **Overview of UnMode.** Given a prompt x sampled from a prompt dataset $\mathcal{D}$, the current policy $p_{\theta^{(r)}}$ generates a response $\hat{y}$, forming the forget set for round r. UnMode then updates the policy via self-distillation: it minimizes the per-token KL divergence between the current model p_θ and a target distribution $\tilde{p}$, which is obtained from the model's own predictions by lowering the logit of each sampled token (see §3.3). This yields $p_{\theta^{(r+1)}}$, from which new responses are sampled to construct the forget set for the next round.

cost. We instead investigate whether diversity can be recovered by removing the dominant response modes that cause repetitive generations.

We adopt *on-policy unlearning*, in which forget targets are sampled from the model's current policy rather than fixed in advance (Jang et al., 2023; Zhang et al., 2024; Dong et al., 2025). At each round, the model generates responses to the input prompts and uses them as forget targets. Unlike CiPO (Li et al., 2026), which constructs counterfactual responses from predefined unlearning targets, and PACT (Tang and Khanna, 2026), which reuses a fixed forget set generated by the original model, our method regenerates forget targets from the evolving policy without predefined targets, coupling target generation with unlearning so that it adapts continuously to the model's current behavior.

We apply this framework to response diversity and introduce UnMode (Figure 1). At each round, UnMode samples responses from the current policy and uses them as forget targets, reducing their probability through parameter updates. Because these targets are regenerated after each update, the method repeatedly suppresses responses that remain likely under the evolving policy, making alternative behaviors relatively more likely. Unlike methods that use auxiliary output heads to suppress previously generated patterns during decoding (Shi and Pan, 2026) or unlearn sampled non-mode responses (Chaitanya et al., 2025), UnMode updates the model using its own on-policy generations as forget targets. It requires no candidate selection, separately sampled alternative responses, or curated replacement data.

Beyond diversity, we show that reducing mode collapse can improve the utility of generated outputs. On Qwen3-4B and Qwen3-8B (Yang et al., 2025), UnMode increases diversity and improves utility scores on the open-ended generation benchmarks Infinity-Chat (Jiang et al., 2025) and NoveltyBench (Zhang et al., 2025b). On HumanEval (Chen et al., 2021) code completion, the increased exploration induced by UnMode improves pass@5 and pass@10, showing that greater response diversity can benefit solution generation.

Our contributions are summarized as follows:

- We formulate *on-policy unlearning*, a general framework where forget targets are dynamically generated from the model's own policy rather than predefined externally.
- We propose UnMode, an on-policy unlearning method for increasing LLM response diversity without requiring sampled alternative responses or curated replacement data.
- We demonstrate that UnMode improves diversity and response utility on open-ended generation benchmarks and increases pass@k on code completion tasks.

2 Related Work

2.1 Training-Time Strategies for Diversity

Post-training can reduce output diversity (Li et al., 2025), motivating diversity-promoting dialogue objectives (Li et al., 2016) and diversity-preserving supervised fine-tuning (Li et al., 2024). DivPO (Lanchantin et al., 2025) favors rare, high-quality responses in preference optimization. In RL, entropy collapse (Cui et al., 2025) has motivated diversity-aware rewards: DARLING (Li et al., 2025) combines semantic diversity with quality, DQO (Chen et al., 2026b) optimizes determinantal-point-process diversity, DIVER (Hu

et al., 2026b) uses sequence-level diversity potentials, and Uniqueness-Aware RL (Hu et al., 2026a) rewards rare correct strategies. Other methods regulate entropy dynamics during policy optimization (Petrenko et al., 2026; Wang et al., 2026). Complementarily, multi-agent approaches elicit diversity through interaction: Multi-Agent Debate (Liang et al., 2024) uses judge-mediated debate to mitigate degeneration of thought, while LLM Discussion (Lu et al., 2024) assigns distinct roles in structured discussions to reduce model homogeneity.

2.2 Test-Time Strategies for Diversity

Test-time methods improve diversity without updating model parameters. Decoding methods reshape sampling via adaptive truncation (top-k (Fan et al., 2018), nucleus (Holtzman et al., 2020), Mirostat (Basu et al., 2021), min-p (Nguyen et al., 2024)), information-theoretic thresholds (epsilon-sampling (Freitag et al., 2023), p-less sampling (Tan et al., 2026)), or penalties for similarity to prior continuations (contrastive search (Su and Collier, 2023), Control-the-Temperature (Troshin et al., 2025)). CreativityNeuro (Schapiro et al., 2026) instead steers selected weights without gradient-based fine-tuning. Prompt-based methods elicit diversity through candidate distributions (Verbalized Sampling (Zhang et al., 2025a)), regeneration conditioned on prior answers (Zhang et al., 2025b), or random-string seeds (SSOT (Misaki and Akiba, 2026)). Finally, Zhang et al. (2026) unify these methods by where diversity is injected and propose generating diverse intermediate specifications to guide final responses.

2.3 Unlearning to Restore Diversity

Machine unlearning aims to remove the influence of a forget set D_f while preserving performance on a retain set D_r (Xu et al., 2024). Approximate parameter-tuning methods include Gradient Ascent (GA) (Jang et al., 2023), which maximizes forget loss but is unstable (Zhang et al., 2024); NPO (Zhang et al., 2024), which uses preference optimization against a static reference; UnDIAL (Dong et al., 2025), which self-distills targets with a fixed logit offset; and Unilogit (Vasilev et al., 2025), which we adopt and computes a logit shift that assigns the target token uniform probability. Recent work has also explored model-generated data for unlearning. CiPO (Li et al., 2026) uses on-policy samples as dispreferred responses but requires separately constructed counterfactual targets based on predefined unlearning targets, while PACT (Tang and Khanna, 2026) uses fixed, model-generated counterfactual examples to unlearn context-triggered deception. In contrast, our framework regenerates forget targets from the evolving policy each round, requiring only prompts.

A growing body of work uses unlearning-style interventions to mitigate mode collapse. Chaitanya et al. (2025) fine-tune on curated non-mode responses, conceptually related to DivPO (Lanchantin et al., 2025). OverRIDE (Shi and Pan, 2026) suppresses previously generated patterns via auxiliary output heads, paralleling approaches that subtract learned forget components from the base model (Ji et al., 2024, 2026). EEPO (Chen et al., 2026a) applies sample-then-forget unlearning during RL rollouts to suppress dominant trajectories. Beyond language generation, Disley and Khan (2026) remove concepts to reduce design fixation in human–AI co-creation, promoting divergent ideation. In contrast, our framework directly unlearns high-probability responses sampled from the current policy during training, without requiring curated diverse responses.

3 Method

This section introduces on-policy unlearning, which dynamically generates forget targets from the model's current policy, and UnMode, its application to mitigating response mode collapse.

3.1 On-Policy Unlearning

Existing unlearning methods typically assume a fixed forget set of input-target pairs (x, y), where the targets are provided externally. We consider an alternative setting where the forget targets are generated from the model's current policy at each round.

Given a prompt set $D = \{x_i\}_{i=1}^n$, at unlearning round r, we sample:

$$\hat{y}_i^{(r)} \sim p_{\theta^{(r-1)}}(y|x_i), \qquad i = 1, \ldots, n, \qquad (1)$$

where $p_{\theta^{(r-1)}}(y \mid x_i)$ denotes the model's conditional generation distribution for round $r-1$, yielding the forget set:

$$D_{\text{forget}}^{(r)} = \{(x_i, \hat{y}_i^{(r)})\}_{i=1}^n. \qquad (2)$$

We refer to this procedure as *on-policy unlearning*: the targets to be suppressed are dynamically

Algorithm 1 On-Policy Unlearning for Response Diversity

Require: Prompt dataset $D = \{x_i\}_{i=1}^{n}$; language model p_θ; rounds R; epochs E

for $1, \ldots, R$ **do**
  Sample $\hat{y}_i \sim p_\theta(\cdot \mid x_i)$ and create $D_{\text{forget}} \leftarrow \{(x_i, \hat{y}_i)\}_{i=1}^{n}$
  for $1, \ldots, E$ **do**
    Compute $\tilde{p}_i$ for each $(x_i, \hat{y}_i) \in D_{\text{forget}}$ via Equation (4)
    Calculate loss $\mathcal{L}_{\text{UNMODE}}(\theta) = \frac{1}{n}\sum_{i=1}^{n} \text{KL}\big(p_\theta \,\|\, \tilde{p}_i\big)(\hat{y}_i \mid x_i)$ and update θ
  end for
end for

generated from the current policy rather than being fixed beforehand. After updating the model using the unlearning loss on $D_{\text{forget}}^{(r)}$, the forget set is regenerated from the updated policy for the next round. This iterative process allows the unlearning targets to continuously adapt to the model's current behavior.

This is motivated by the observation that a given prompt can elicit multiple valid generations from a model, allowing on-policy targets to capture diverse realizations of the model's behavior rather than a single predefined response.

3.2 UNMODE: On-Policy Unlearning for Response Diversity

We apply on-policy unlearning to reduce response mode collapse. For a prompt x, let M denote a semantic response mode with probability mass

$$p_m = p_\theta(y \in M \mid x).$$

When responses are sampled from the current policy, modes with greater probability mass are more likely to be represented in the forget set. UNMODE uses these sampled responses as forget targets, reducing the probability of generating them again. After each update, it samples new responses from the updated policy. This process repeatedly targets responses that remain likely under the evolving policy, making other responses relatively more likely.

3.3 UNMODE Loss

To implement the UNMODE loss, we use Unilogit (Vasilev et al., 2025), which performs self-distillation toward a modified target distribution. Given a sampled response $\hat{y}_i^{(r)}$, let $v^\star = \hat{y}_{i,t}^{(r)}$ be the token to suppress at position t. Unilogit modifies the logits as:

$$\tilde{z}(x) = (1 - \mathbf{1}_{v^\star})z(x) + \mathbf{1}_{v^\star} \log \sum_{k \neq v^\star} \frac{\exp(z_k(x))}{V - 1}, \tag{3}$$

where $\mathbf{1}_{v^\star}$ is a one-hot vector indicating the selected token, and V is the vocabulary size. The target distribution is:

$$\tilde{p}_i(\cdot|x_i) = \text{softmax}(\tilde{z}(x_i)). \tag{4}$$

The UNMODE loss minimizes the KL divergence between the modified target and the current model distribution:

$$\mathcal{L}_{\text{UNMODE}}^{(r)}(\theta) = \frac{1}{n}\sum_{i=1}^{n} \text{KL}\left(p_\theta \| \tilde{p}_i\right)(\hat{y}_i^{(r)}|x_i). \tag{5}$$

By iteratively applying this objective to responses sampled from the current policy, UNMODE decreases the probability of repeatedly generating the same responses, making alternative responses relatively more likely. The complete procedure is summarized in Algorithm 1.

4 Experimental Setup and Results

4.1 Setup

Models. We evaluate two model sizes from the Qwen3 family, Qwen3-4B and Qwen3-8B (Yang et al., 2025).

Baselines. We compare UNMODE against three baselines, each chosen for a specific purpose: (i) *Direct*, which samples directly from the base model without any intervention and serves as the reference for measuring how much our method improves diversity and utility; (ii) **DivPO** (Lanchantin et al., 2025), a preference-optimization method that trains on the most diverse high-quality responses as preferred examples, representing an effective post-training approach that requires a candidate

pool to construct a diverse training dataset; and (iii) **OverRIDE** (Shi and Pan, 2026), a decoding-time method that improves diversity through iterative reweighting, representing an effective approach that avoids constructing a candidate pool by steering generation away from previously generated responses.

To compare DivPO under both our experimental setting and the original DivPO setting, we evaluate two variants on open-ended generation: (i) DivPO, trained on the same prompt dataset as all other methods, and (ii) DivPO-5k, trained on a random sample of 5k prompts from Infinity-Chat (Jiang et al., 2025) to match the scale of the training data used in the original DivPO paper.

Implementation Details. All experiments use LoRA (Hu et al., 2022) with a rank of 16 and an alpha of 32 for fine-tuning, and vLLM (Kwon et al., 2023) for response generation. Unless otherwise specified, the sampling temperature is set to 1.0. Each experiment is run with three random seeds, and we report the mean and standard deviation across seeds. Further implementation details are provided in Appendix A.

4.1.1 Open-Ended Generation

Datasets. We evaluate our method on two open-ended benchmarks: (i) INFINITY-CHAT100 (Jiang et al., 2025), which was introduced to study intra- and inter-model repetition, and (ii) the NB-CURATED subset of NOVELTYBENCH (Zhang et al., 2025b). Both contain 100 prompts but capture open-endedness from complementary perspectives. INFINITY-CHAT100 is human-verified to ensure that its prompts admit open-ended responses (Jiang et al., 2025), while NB-CURATED spans four categories: randomness, factual questions with multiple valid answers, creative writing, and subjectivity. All experiments use cross-dataset evaluation in both directions: we train on one benchmark and test on the other. Henceforth, we refer to INFINITY-CHAT100 and NB-CURATED as Infinity-Chat and NoveltyBench, respectively.

Evaluation Metrics. For each prompt, we sample 10 responses to evaluate response diversity and utility. *Self-BLEU* measures lexical overlap among responses, and average pairwise embedding similarity (OpenAI's `text-embedding-3-small`) measures semantic overlap; lower values indicate greater diversity. Following NoveltyBench (Zhang et al., 2025b), we also report two metrics defined over functional equivalence classes, where responses are grouped using a fine-tuned `DeBERTa-v3-large` (He et al., 2021) classifier that judges whether a user would gain anything from seeing one response after the other. Let c_i be the class of the i-th response. *Distinct* is the number of distinct classes, $|\{c_i\}_{i=1}^{k}|$. *Utility* combines novelty and quality,

$$\text{Utility}_k = \frac{1-p}{1-p^k} \sum_{i=1}^{k} p^{i-1} \cdot \mathbb{1}[c_i \neq c_j, \forall j < i] \cdot u_i,$$

where u_i denotes the quality score of response i, and $p = 0.8$ is a patience factor that geometrically discounts later responses; responses equivalent to earlier ones contribute zero. Following NoveltyBench's protocol, we score response quality using `Skywork-Reward-Gemma-2-27B-v0.2` (Liu et al., 2024). NoveltyBench selects this model based on its strong RewardBench performance (Lambert et al., 2025) and reported superiority to larger models, including NVIDIA's 70B Nemotron model (Wang et al., 2025).

4.1.2 Code Completion

Datasets. We evaluate on HUMANEVAL (Chen et al., 2021). For training, we use 100 prompts sampled from MBPP (Austin et al., 2021), which we convert into a code-completion format to match the HUMANEVAL setting. For each problem, we construct a function signature from the function name in the reference solution and use the original MBPP task description as the docstring, with the model asked to complete the function body. Since the training and evaluation problems come from different datasets, this setup also evaluates cross-benchmark transfer of the resulting improvements.

Evaluation Metrics. For each problem, we sample 100 completions and report the unbiased Pass@k estimator (Chen et al., 2021) for $k \in \{1, 5, 10\}$. To measure diversity among the generated programs, we additionally report average pairwise embedding similarity, computed using embeddings from OpenAI's `text-embedding-3-small` API.

4.2 Results.

Open-Ended Generation. Table 1 reports results on Infinity-Chat and NoveltyBench.

UNMODE improves both diversity and utility over the base model. On Qwen3-8B, UNMODE

Method	Infinity-Chat				NoveltyBench			
	Self-BLEU ↓	Sim ↓	Distinct↑	Utility ↑	Self-BLEU ↓	Sim ↓	Distinct ↑	Utility ↑
Qwen3-8B								
Direct	$.64_{\pm.01}$	$.87_{\pm.00}$	$2.52_{\pm.12}$	$2.91_{\pm.02}$	$.61_{\pm.01}$	$.82_{\pm.00}$	$3.84_{\pm.11}$	$3.43_{\pm.04}$
OverRIDE	$.63_{\pm.01}$	$.86_{\pm.00}$	$2.61_{\pm.02}$	$2.96_{\pm.07}$	$.60_{\pm.00}$	$.81_{\pm.01}$	$4.01_{\pm.08}$	$3.54_{\pm.03}$
DivPO	$.59_{\pm.01}$	$.84_{\pm.00}$	$2.84_{\pm.06}$	$3.14_{\pm.05}$	$.55_{\pm.01}$	$.80_{\pm.00}$	$4.24_{\pm.12}$	$3.61_{\pm.07}$
DivPO-5k	$.57_{\pm.01}$	$.84_{\pm.00}$	$3.07_{\pm0.08}$	$3.27_{\pm0.05}$	$.54_{\pm.01}$	$.79_{\pm.00}$	$4.73_{\pm.18}$	$3.91_{\pm.04}$
UNMODE	$\mathbf{.40}_{\pm.00}$	$\mathbf{.78}_{\pm.00}$	$\mathbf{4.79}_{\pm.15}$	$\mathbf{4.27}_{\pm.21}$	$\mathbf{.34}_{\pm.00}$	$\mathbf{.68}_{\pm.01}$	$\mathbf{7.69}_{\pm.08}$	$\mathbf{4.64}_{\pm.05}$
Qwen3-4B								
Direct	$.65_{\pm.00}$	$.86_{\pm.00}$	$2.60_{\pm.06}$	$2.88_{\pm.07}$	$.63_{\pm.00}$	$.83_{\pm.00}$	$3.78_{\pm.10}$	$3.09_{\pm.09}$
OverRIDE	$.63_{\pm.01}$	$.86_{\pm.00}$	$2.71_{\pm.10}$	$2.94_{\pm.07}$	$.61_{\pm.01}$	$.82_{\pm.00}$	$3.94_{\pm.13}$	$3.19_{\pm.05}$
DivPO	$.63_{\pm.00}$	$.85_{\pm.00}$	$2.67_{\pm.15}$	$2.90_{\pm.06}$	$.61_{\pm.00}$	$.82_{\pm.00}$	$3.79_{\pm.14}$	$3.19_{\pm.07}$
DivPO-5k	$.59_{\pm.00}$	$\mathbf{.84}_{\pm.00}$	$2.85_{\pm.05}$	$3.02_{\pm.10}$	$.58_{\pm.00}$	$.81_{\pm.00}$	$4.42_{\pm.07}$	$\mathbf{3.42}_{\pm.08}$
UNMODE	$\mathbf{.57}_{\pm.00}$	$.85_{\pm.00}$	$\mathbf{3.26}_{\pm.11}$	$\mathbf{3.12}_{\pm.14}$	$\mathbf{.45}_{\pm.00}$	$\mathbf{.76}_{\pm.00}$	$\mathbf{6.39}_{\pm.12}$	$\mathbf{3.42}_{\pm.14}$

Table 1: Main results on Infinity-Chat and NoveltyBench for Qwen3-8B and Qwen3-4B. We report Self-BLEU, embedding similarity (Sim), number of distinct responses, and Utility. UNMODE improves diversity and utility over the base model (Direct), outperforms OverRIDE and DivPO, and matches or exceeds DivPO-5k, which requires a far more expensive response-pool construction.

lowers Self-BLEU from .64 to .40 on Infinity-Chat and from .61 to .34 on NoveltyBench, and nearly doubles the number of distinct responses (from 2.52 to 4.79 and 3.84 to 7.69, respectively). Utility also increases (from 2.91 to 4.27 on Infinity-Chat and from 3.43 to 4.64 on NoveltyBench), and Qwen3-4B shows the same trend, indicating that the observed diversity gains do not come at the expense of utility. Lowering the probability of the model's own frequent responses pushes it toward alternative answers that remain correct and useful, which is what open-ended prompts call for.

UNMODE outperforms OverRIDE and DivPO. OverRIDE yields only small gains over the base model. DivPO helps on Qwen3-8B but is much less effective on Qwen3-4B, where it is nearly indistinguishable from the base model. UNMODE gives the best diversity and utility in nearly all settings for both model sizes.

UNMODE matches or exceeds DivPO-5k at a much lower cost. DivPO-5k is the strongest baseline, but it requires building a candidate pool of 100 responses for each of 5k prompts before training. On Qwen3-8B, UNMODE clearly outperforms it on all metrics (e.g., 4.79 vs. 3.07 distinct responses and 4.27 vs. 3.27 utility on Infinity-Chat). On Qwen3-4B, UNMODE is better on NoveltyBench (6.39 vs. 4.42 distinct responses, with equal utility) and comparable on Infinity-Chat, where it is ahead on Self-BLEU, Distinct, and Utility, while DivPO-5k is slightly ahead on Sim. OverRIDE, DivPO, and UNMODE share the same premise: a model already assigns some probability to less likely but valid and diverse responses, and these must be brought out of the model's distribution. OverRIDE does this at decoding time, and DivPO does it by training on a curated pool of preference pairs. UNMODE instead only reduces the probability of responses the model itself generates, and it does not need to construct or rank a candidate pool. This simple on-policy update surfaces diverse responses more effectively than the baselines and at a lower cost than DivPO.

Method	Sim ↓	P@1	P@5	P@10
Qwen3-8B				
Direct	$.94_{\pm.0}$	$60.1_{\pm0.1}$	$71.4_{\pm0.2}$	$75.1_{\pm0.3}$
OverRIDE	$.93_{\pm.0}$	$60.1_{\pm0.3}$	$72.9_{\pm0.8}$	$77.0_{\pm0.9}$
DivPO	$.94_{\pm.0}$	$54.6_{\pm0.1}$	$66.9_{\pm0.2}$	$70.4_{\pm0.3}$
UNMODE	$.90_{\pm.0}$	$57.4_{\pm0.2}$	$\mathbf{77.0}_{\pm0.2}$	$\mathbf{82.0}_{\pm0.1}$
Qwen3-4B				
Direct	$.94_{\pm.0}$	$32.2_{\pm0.1}$	$40.3_{\pm0.6}$	$43.2_{\pm0.9}$
OverRIDE	$.94_{\pm.0}$	$32.4_{\pm0.3}$	$41.6_{\pm0.4}$	$44.9_{\pm0.4}$
DivPO	$.95_{\pm.0}$	$33.1_{\pm0.5}$	$40.7_{\pm0.9}$	$43.5_{\pm0.9}$
UNMODE	$.94_{\pm.0}$	$34.5_{\pm0.6}$	$\mathbf{45.5}_{\pm0.6}$	$\mathbf{49.3}_{\pm0.3}$

Table 2: Code completion results on HumanEval for Qwen3-8B and Qwen3-4B. Sim is the embedding similarity between generated solutions, and P@k denotes pass@k, the percentage of problems solved by at least one of k sampled solutions. UNMODE achieves the highest pass@5 and pass@10, showing that it surfaces correct alternative solutions, at a small pass@1 cost on Qwen3-8B.

Method	Infinity-Chat				NoveltyBench			
	Self-BLEU ↓	Sim ↓	Distinct ↑	Utility ↑	Self-BLEU ↓	Sim ↓	Distinct ↑	Utility ↑
Qwen3-8B								
Direct	$.64_{\pm.01}$	$.87_{\pm.00}$	$2.52_{\pm.12}$	$2.91_{\pm.02}$	$.61_{\pm.01}$	$.82_{\pm.00}$	$3.84_{\pm.11}$	$3.43_{\pm.04}$
UNMODE (off-policy)	$.35_{\pm.00}$	$\mathbf{.77}_{\pm.00}$	$\mathbf{6.00}_{\pm.05}$	$3.89_{\pm.10}$	$.50_{\pm.00}$	$.77_{\pm.00}$	$5.67_{\pm.11}$	$4.01_{\pm.07}$
UNMODE (on-policy)	$\mathbf{.34}_{\pm.00}$	$\mathbf{.77}_{\pm.00}$	$5.87_{\pm.23}$	$\mathbf{4.45}_{\pm.09}$	$\mathbf{.44}_{\pm.00}$	$\mathbf{.75}_{\pm.00}$	$\mathbf{6.63}_{\pm.10}$	$\mathbf{4.07}_{\pm.06}$
Qwen3-4B								
Direct	$.65_{\pm.00}$	$.86_{\pm.00}$	$2.60_{\pm.06}$	$2.88_{\pm.07}$	$.63_{\pm.00}$	$.83_{\pm.00}$	$3.78_{\pm.10}$	$3.09_{\pm.09}$
UNMODE (off-policy)	$.56_{\pm.00}$	$.84_{\pm.00}$	$3.20_{\pm.09}$	$3.02_{\pm.09}$	$.62_{\pm.00}$	$.82_{\pm.00}$	$3.93_{\pm.03}$	$3.20_{\pm.01}$
UNMODE (on-policy)	$\mathbf{.37}_{\pm.00}$	$\mathbf{.78}_{\pm.00}$	$\mathbf{6.12}_{\pm.05}$	$\mathbf{3.46}_{\pm.10}$	$\mathbf{.56}_{\pm.00}$	$\mathbf{.80}_{\pm.00}$	$\mathbf{4.79}_{\pm.18}$	$\mathbf{3.27}_{\pm.14}$

Table 3: Test-time UNMODE results on Infinity-Chat and NoveltyBench. Direct denotes generation without test-time unlearning. The off-policy variant reuses a fixed forget set, whereas the on-policy variant regenerates the forget target after each update.

Code Completion. Table 2 reports results on HumanEval, where we measure diversity with the embedding similarity of generated solutions and correctness with pass@k for $k \in \{1, 5, 10\}$.

UNMODE finds correct solutions that the baselines miss. UNMODE achieves the highest pass@5 and pass@10 for both model sizes. On Qwen3-8B, pass@10 rises from 75.1 (Direct) to 82.0, and from 77.0 for OverRIDE, the strongest baseline. On Qwen3-4B, it rises from 43.2 to 49.3. The gains are also visible in similarity: UNMODE produces less similar solutions on Qwen3-8B (.90 vs. .94), and on Qwen3-4B it matches the baselines. Since pass@k increases with the number of samples only if those samples differ from one another, the improvement at larger k indicates that UNMODE reduces redundant generations and explores a wider set of candidate solutions, which raises the chance that at least one of them is correct.

Diversity comes with a small cost at pass@1 on Qwen3-8B. On Qwen3-8B, pass@1 drops from 60.1 (Direct) to 57.4. This is expected: pass@1 rewards the single sampled solution, and spreading probability mass over alternative solutions can slightly lower the chance that any one sample is correct. This cost is outweighed once several samples are drawn, as the pass@5 and pass@10 gains (+5.6 and +6.9 points over Direct) show. On Qwen3-4B, pass@1 does not decrease and instead improves (32.2 to 34.5). DivPO shows a larger pass@1 drop on Qwen3-8B without comparable gains at higher k, so it does not offer this trade-off.

5 Analysis

5.1 Test-Time UnMode

Although we primarily study UNMODE as a post-training procedure, it can also be applied at test time to diversify responses to individual prompts. For each prompt, we iteratively generate a response from the current model and use it as the forget target for an unlearning update, with each update performed specifically for that prompt. After r rounds, we sample n responses from the resulting prompt-specific model and evaluate their diversity and utility. Thus, the evaluated responses are sampled from the final updated model rather than collected one per unlearning round.

We evaluate this test-time procedure on Infinity-Chat and NoveltyBench, comparing it with direct generation and an off-policy variant that reuses a fixed set of responses rather than regenerating the forget target after each update. As shown in Table 3, test-time UNMODE improves response diversity and utility over direct generation across both model sizes and datasets. Moreover, the on-policy variant generally achieves better results than its off-policy counterpart, highlighting the benefit of regenerating forget targets as the model's response distribution changes. These findings demonstrate that UNMODE can be used not only as a post-training procedure but also as an iterative, prompt-specific intervention at inference time.

5.2 Diversity–Utility Trade-off

Figure 2 illustrates the trade-off between response diversity and utility as the number of unlearning rounds R increases. Across model sizes, additional unlearning rounds generally improve diversity, but excessive unlearning can eventually reduce utility.

5.3 Evolution of Unlearning Targets

We examine whether regenerating unlearning targets throughout training yields a more diverse set of responses than sampling targets once from the initial model. We compare two strategies: *on-policy*,

Method	Infinity-Chat				NoveltyBench			
	Self-BLEU ↓	Sim ↓	Distinct ↑	Utility ↑	Self-BLEU ↓	Sim ↓	Distinct ↑	Utility ↑
Qwen3-8B								
Off-policy	$.64_{\pm.01}$	$.87_{\pm.00}$	$2.51_{\pm.09}$	$2.90_{\pm.03}$	$.62_{\pm.01}$	$.82_{\pm.00}$	$3.84_{\pm.11}$	$3.43_{\pm.04}$
On-policy	$\mathbf{.46}_{\pm.01}$	$\mathbf{.78}_{\pm.00}$	$\mathbf{4.20}_{\pm.01}$	$\mathbf{3.38}_{\pm.03}$	$\mathbf{.46}_{\pm.01}$	$\mathbf{.72}_{\pm.01}$	$\mathbf{6.02}_{\pm.11}$	$\mathbf{3.83}_{\pm.10}$
Qwen3-4B								
Off-policy	$.65_{\pm.00}$	$.86_{\pm.00}$	$2.60_{\pm.06}$	$2.88_{\pm.07}$	$.63_{\pm.00}$	$.83_{\pm.00}$	$3.78_{\pm.10}$	$3.09_{\pm.09}$
On-policy	$\mathbf{.56}_{\pm.00}$	$\mathbf{.81}_{\pm.00}$	$\mathbf{3.48}_{\pm.03}$	$\mathbf{3.01}_{\pm.03}$	$\mathbf{.52}_{\pm.01}$	$\mathbf{.76}_{\pm.00}$	$\mathbf{5.35}_{\pm.10}$	$\mathbf{3.20}_{\pm.06}$

Table 4: Diversity and utility of unlearning target sets obtained using on-policy and off-policy sampling on Infinity-Chat and Novelty Bench. On-policy targets are regenerated across unlearning rounds, whereas off-policy targets are sampled from the initial model and reused throughout training.

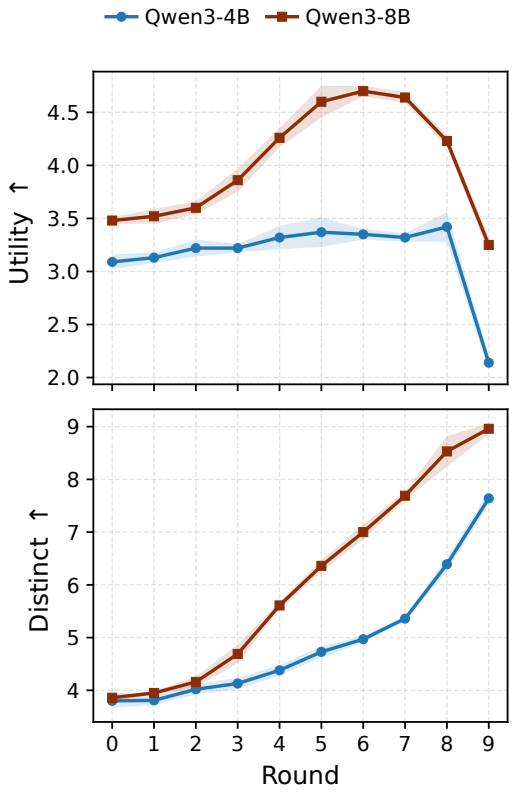


Figure 2: Response diversity and utility across unlearning rounds on NoveltyBench.

which samples one response per prompt from the initial model and then from the current model at each of the nine subsequent rounds, yielding ten responses per prompt; and *off-policy*, which samples all ten responses per prompt once from the fixed initial model. Since the model's response distribution evolves during unlearning, on-policy sampling can introduce responses that were less likely under earlier model distributions, whereas the fixed off-policy set remains restricted to responses sampled initially.

To assess the resulting target sets, we evaluate their diversity and utility on Infinity-Chat and NoveltyBench. As shown in Table 4, on-policy targets exhibit lower Self-BLEU and similarity, as well as higher Distinct and Utility scores, across both model sizes and datasets. These results indicate that periodically regenerating targets yields a more diverse set of responses than a fixed set sampled from the initial model. As unlearning lowers the probability of previously generated responses, the model shifts toward semantically different ones, which are then incorporated into subsequent unlearning targets, whereas the off-policy set remains limited to the initial model's responses.

5.4 Effect of Sampling Temperature

We study how sampling temperature (T) affects UnMode during inference and during training, when sampling the responses used to construct the forget set.

Inference temperature. Figure 3 reports results for a model trained at temperature 0.7 and evaluated at inference temperatures from 0.5 to 1.0. On NoveltyBench, higher temperature improves utility for every method, but UnMode achieves the highest utility at every temperature, and its advantage widens as temperature increases. At T=0.5 it leads the strongest baseline (DivPO-5k) by 0.56 utility points, and at T=1.0 by 1.13. On HumanEval, Pass@10 also rises with temperature for all methods. UnMode again scores highest throughout, reaching 81.6 at T=1.0 against 77.0 for OverRIDE, 75.3 for Direct, and 73.0 for DivPO. These results show that UnMode's advantage holds across inference temperatures and grows as temperature increases, indicating that its gains do not depend on a particular inference temperature.

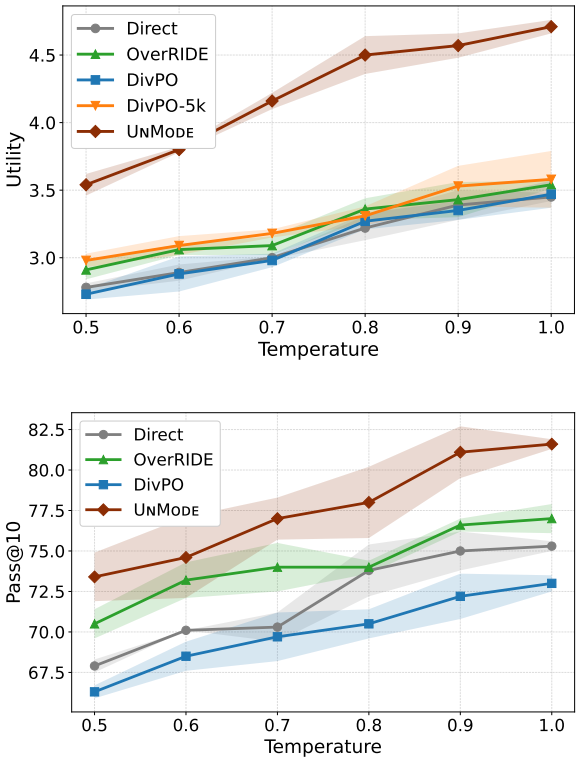


Figure 3: Effect of inference temperature on NoveltyBench utility (top) and HumanEval pass@10 (bottom). All models are trained at temperature 0.7 and evaluated at inference temperatures from 0.5 to 1.0. Shaded regions denote standard deviation.

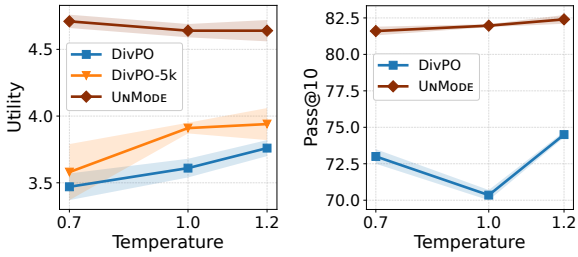


Figure 4: Effect of training temperature on NoveltyBench utility (left) and HumanEval pass@10 (right). Models are trained with data sampled at temperatures 0.7, 1.0, and 1.2 and evaluated at inference temperature 1.0. Shaded regions denote standard deviation.

Training temperature. Figure 4 varies the temperature used to sample training data over $\{0.7, 1.0, 1.2\}$, with evaluation fixed at T=1.0. UnMode is largely insensitive to this choice. Utility on NoveltyBench changes only slightly, and Pass@10 on HumanEval stays within one point (81.6 to 82.4). DivPO is more sensitive. Its utility rises with training temperature, and its Pass@10 is non-monotonic. These results suggest that UnMode does not depend on careful tuning of the training temperature.

6 Conclusion

We studied whether response diversity in open-ended generation can be recovered by reducing the probability of responses that a model repeatedly produces. To this end, we adopted on-policy unlearning: at each round, we sample responses from the current model and unlearn them, so the forget targets follow the model's evolving output distribution. The resulting method, UnMode, requires neither a curated pool of preferred responses nor predefined unlearning targets. On Qwen3-4B and Qwen3-8B, trained on one open-ended benchmark and evaluated on the other, UnMode improved diversity and utility over the base model, outperformed base models, OverRIDE, and DivPO baselines, and matched or exceeded DivPO-5k, which relies on a substantially more expensive response-pool construction. On HumanEval, UnMode improved pass@5 and pass@10 for both model sizes, at a small pass@1 cost on Qwen3-8B. Our analysis indicated that on-policy targets are more diverse than targets sampled once. We also observed that excessive unlearning eventually reduces utility, so the number of rounds needs to be chosen with care.

7 Limitations

Our study has several limitations. We evaluate only the Qwen3 family at two model sizes, and we use relatively small training sets. The gains are also smaller on Qwen3-4B than on Qwen3-8B for some metrics. Future work could test whether the findings hold across other model families and scales, and could examine how on-policy unlearning interacts with other post-training objectives.

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A Experimental Details

Tables 5, 6, and 7 report the training configurations for UnMode, OverRIDE, and DivPO, respectively. For DivPO, we use the same reward model as the original paper, ArmoRM-Llama3-8B-v0.1 (Wang et al., 2024), to score candidate responses and construct preference pairs. Table 8 lists the evaluation parameters used for all methods.

Checkpoint selection. For open-ended generation tasks, we select the round checkpoint that achieves the highest utility score on the training data. For code generation, we use the final checkpoint, i.e., the model after the fixed number of unlearning rounds.

Parameter	Training-time	Test-time
Learning Rate	5×10^{-7}	5×10^{-5}
Effective Batch Size	1	1
LoRA Rank (r)	16	16
LoRA Alpha (α)	32	32
LoRA Target Modules	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj	
Sampling Temperature	1.0	1.0
Epochs per Round	2	2
Max Rounds	9	9

Table 5: Training Configuration for Training-time and Test-time UNMODE

Parameter	OverRIDE
Learning Rate	1×10^{-3}
LoRA Rank (r)	16
LoRA Alpha (α)	32
LoRA Target Modules	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
Reweighting Parameter (λ)	0.8
Rounds (T)	10
Sampling Temperature	1.0

Table 6: Training Configuration for OverRIDE.

Parameter	DivPO
Learning Rate	1×10^{-6}
Effective Batch Size	16
LoRA Rank (r)	16
LoRA Alpha (α)	32
LoRA Target Modules	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
Reward Threshold (ρ)	0.3
Diversity Criterion (D)	Prob
Sampling Temperature	1.0
Number of Training Steps	1000

Table 7: Training Configuration for DivPO.

Parameter	**Open-Ended**	**Code**
Max New Tokens	256	512
Samples per Prompt	10	100
Thinking Mode	Disabled	Disabled
Temperature	1.0	1.0
Top-p	0.95	0.95
Top-k	0	0
Min-p	0.0	0.0

Table 8: Evaluation parameters used for open-ended generation and code completion.